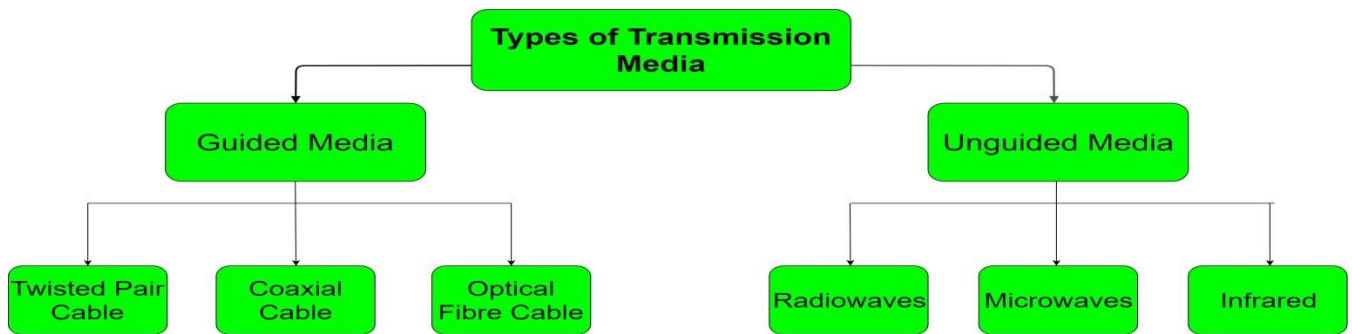




Media	Working principle	Advantage	Disadvantage	Who used?
Twisted Pair Cable	It consists of 2 separately insulated conductor wires wound about each other. Generally, several such pairs are bundled together in a protective sheath. They are the most widely used Transmission Media. When you use the twisted pair, the noise source is equally close to each of the wires. Therefore, the two wires pick up roughly equal noise voltages with the same polarity. Since the current running in one wire is equal but opposite of the other wire, the two magnetic fields cancel each other out!	1. It can be used to carry both analog and digital data. 2. It is relatively easy to implement and terminate. 3. It is the least expensive media of transmission for short distances. 4. If portion of a twisted pair cable is damaged it does not effect the entire network.	1. It offers poor noise immunity as a result signal distortion is more? 2. Attenuation is very high. 3. It supports lower bandwidth as compared to other Medias. It supports 10 mbps upto a distance of 100 meters on a 10BASE-T. 4. It offers very poor security and is relatively easy to tap. 5. Being thin in size, they are likely to break easily	(1) In telephone lines to carry voice and data channels. (2) In the local loop. (3) In the DSL line (ADSL) (4) Local area networks such as 10 Base-T and 100 Base-T. Use the twisted pair cables. (5) In the ISDN (Integrate d Services Digital Network).

Media	Working principle	Advantage	Disadvantage	Who used?
Unshielded twisted pair (UTP)	Inside a UTP cable is up to four twisted pairs of copper wires, enclosed in a protective plastic cover, with the greater number of pairs corresponding to more bandwidth. The two individual wires in a single pair are twisted around each other, and then the pairs are twisted around each other, as well. This is done to reduce crosstalk and electromagnetic interference, each of which can degrade network performance. Each signal on a twisted pair requires both wires.	1. Least expensive 2. Easy to install 3. High speed capacity 4. Susceptible to external interference 5. Lower capacity and performance in comparison to STP 6. Short distance transmission due to attenuation	It is more susceptible to interference compare to most of the other cable types. Twisting of pair helps to certain extent but it does not make cable impervious to electrical noise completely. ➤It can be used up to cable segment lengths of about 100 meters only. ➤UTP cable should follow specifications for number of twists or braids permitted per meter of cable to reduce crosstalk.	UTP cables are found in many ethernet network , telephone system , security camera system . Twisted-pair cables are used in the telephone lines to provide voice and data channels.
Shielded Twisted Pair (STP)	• STP is made up of pairs of copper wires that are twisted together. • The pairs are covered in a foil or braided mesh, as well as outer PVC jacket. • This foil or mesh prevents the penetration of electromagnetic noise and eliminate cross talk. • This shielding must be grounded to prevent the foil or braided mesh from becoming a magnet for electricity.	• Better performance at a higher data rate in comparison to UTP • Eliminates crosstalk • Comparitively faster • Comparitively difficult to install and manufacture • More expensive • Bulky	1. It must be properly grounded. 2. It is more expensive than UTP. 3. It is difficult to terminate.	One common use for shielded twisted pair cables is to use it in extreme cold situations. The best use for STP cables are in industrial settings with high amounts of electromagnetic interference, such as a factory with large electronic equipment, where they can be properly installed and maintained.

Media	Working principle	Advantage	Disadvantage	Who used?
Coaxial Cable	The signal a coaxial cable carries is transferred simultaneously through the central wire as well as the separated metal jacketing. This is done because both conductors generate a magnetic field, as any electrically charged wire does. However, when two opposite charged magnetic fields, as the ones generated by the two conductors are, come into contact with one another, the fields cancel one another out. This allows the cables to be placed near other sensitive electronic equipment and other metal objects without the danger of the cables acting like magnets. It also prevents outside magnetic fields from altering the signal the cables carry.	• High Bandwidth • Better noise Immunity • Easy to install and expand • Inexpensive	• Single cable failure can disrupt the entire network	• Wireless and antenna application(connecting from dish to television through coaxial cable. example: Tata sky) • RF and Microwave transmission • Video distribution • Cable television and cable internet
Optical Fibre Cable	It uses the concept of reflection of light through a core made up of glass or plastic. The core is surrounded by a less dense glass or plastic covering called the cladding. It is used for transmission of large volumes of data. The cable can be unidirectional or bidirectional. The WDM (Wavelength Division Multiplexer) supports two modes, namely unidirectional and bidirectional mode.	• Increased capacity and bandwidth • Light weight • Less signal attenuation • Immunity to electromagnetic interference • Resistance to corrosive materials	• Difficult to install and maintain • High cost • Fragile	• Medical Industry • Communication • Defence • Industries • Broadcasting • Lighting and Decorations • Mechanical Inspections

Media	Working principle	Advantage	Disadvantage	Who used?
Radio waves	These are easy to generate and can penetrate through buildings. The sending and receiving antennas need not be aligned. Frequency Range: 3KHz – 1GHz.	Some are reflected off the ionosphere, so can travel around the Earth. Can carry a message instantaneously over a wide area. Aerials to receive them are simpler than for microwaves.	The range of frequencies that can be accessed by existing technology is limited, so there is a lot of competition amongst companies for the use of the frequencies.	They are used in standard broadcast radio and television, shortwave radio, navigation and air-traffic control, cellular telephony, and even remote-controlled toys.
Micro waves	It is a line of sight transmission i.e. the sending and receiving antennas need to be properly aligned with each other. The distance covered by the signal is directly proportional to the height of the antenna. Frequency Range: 1GHz – 300GHz.	Pass through the ionosphere, so are suitable for satellite to Earth transmission. Can be modified to carry many signals at one time, including data, television pictures and voice messages.	Absorbed very easily by natural, e.g. rain, and made objects, e.g. concrete. They are also absorbed by living tissue and may cause harm by their cooking effect. Need special aerials to receive them.	These are majorly used for mobile phone communication and television distribution.
Infrared	Infrared waves are used for very short distance communication. They cannot penetrate through obstacles. This prevents interference between systems. Frequency Range: 300GHz – 400THz.	➡It provides secured communication due to line of sight or point-to-point mode of communication. ➡The battery used in infrared devices last for long duration due to lower power consumption. ➡Infrared motion sensors detect	➡Infrared frequencies are affected by hard objects (e.g. walls, doors), smoke, dust, fog, sunlight etc. Hence it does not work through walls or doors. ➡Infrared waves at high power can damage eyes. ➡In monitor & control	It is used in TV remotes, wireless mouse, keyboard, printer, etc.

		motion in daytime and nighttime reliably. ➡The sensor does not require any contact with the product to be sensed. The infrared devices are more appropriate for targets which are close than 10 mm. ➡Infrared devices can measure distance to soft objects which may not be easily detected by ultrasound. ➡They are physically smaller in size and are more affordable. ➡It has response time faster than thermocouple. ➡It provides good stability over time.	application, it can control only one device at one time. Moreover it is difficult to control things which are not in LOS (Line of Sight). It requires line of sight between transmitter and receiver to communicate. ➡It supports shorter range and hence its performance degrades with longer distances. ➡It supports lower data rate transmission compared to wired transmission.	
--	--	--	---	--